

Algebra II with Trigonometry

Chapter 6.4

Olympiad Solutions (Gemini)

Saved tutoring response with MathJax rendering. Original PDF file:
[4533650_alg-2trig-h-chp-6-4-note.pdf](https://www.gemini.com/4533650_alg-2trig-h-chp-6-4-note.pdf)

Solutions

1. Suppose x is an angle such that $\sec(2024x) + \tan(2024x) = 7$. Find the exact value of $\sec(2024x) - \tan(2024x)$.

Recall the fundamental trigonometric identity $\sec^2 \theta - \tan^2 \theta = 1$. This can be factored as a difference of squares:

$$(\sec \theta - \tan \theta)(\sec \theta + \tan \theta) = 1$$

Substituting $\theta = 2024x$ into the identity, we have:

$$(\sec(2024x) - \tan(2024x))(7) = 1$$

Dividing both sides by 7, the exact value is:

$$\sec(2024x) - \tan(2024x) = \frac{1}{7}$$

2. Compute the exact value of $\tan\left(\frac{\pi}{12}\right) + \cot\left(\frac{\pi}{12}\right)$.

For any angle θ where both functions are defined, we can rewrite the expression in terms of sine and cosine:

$$\tan \theta + \cot \theta = \frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta} = \frac{\sin^2 \theta + \cos^2 \theta}{\sin \theta \cos \theta}$$

By the Pythagorean identity $\sin^2 \theta + \cos^2 \theta = 1$ and the double-angle formula $\sin(2\theta) = 2 \sin \theta \cos \theta$, this simplifies beautifully to:

$$\frac{1}{\sin \theta \cos \theta} = \frac{2}{\sin(2\theta)}$$

Letting $\theta = \frac{\pi}{12}$, we obtain:

$$\tan\left(\frac{\pi}{12}\right) + \cot\left(\frac{\pi}{12}\right) = \frac{2}{\sin\left(\frac{\pi}{6}\right)} = \frac{2}{1/2} = 4$$

3. Evaluate the sum of all x -values strictly between 0 and 2π where the graph of $y = \sec(8x)$ has a vertical asymptote.

Vertical asymptotes of $\sec(8x) = \frac{1}{\cos(8x)}$ occur precisely when $\cos(8x) = 0$. This happens when the argument is an odd multiple of $\frac{\pi}{2}$:

$$8x = \frac{\pi}{2} + k\pi \implies x = \frac{\pi}{16} + \frac{k\pi}{8} = \frac{(2k+1)\pi}{16}$$

for any integer k . We restrict $x \in (0, 2\pi)$, which gives $0 < \frac{2k+1}{16} < 2$, meaning $0 < 2k+1 < 32$. The valid integers are $k = 0, 1, 2, \dots, 15$, yielding 16 distinct solutions.

Since the solutions are roots of an evenly periodic function on a full interval $(0, 2\pi)$, they are symmetric about the midpoint of the interval, which is π . Thus, the sum of the roots is simply the number of roots multiplied by the axis of symmetry:

$$\text{Sum} = 16 \times \pi = 16\pi$$

4. Determine the sum of the first ten positive x -intercepts of the graph of the function $f(x) = \cot\left(\pi x - \frac{\pi}{4}\right)$.

The x -intercepts of the cotangent function occur when its argument is an odd multiple of $\frac{\pi}{2}$ (where cosine is 0 and sine is non-zero), meaning:

$$\pi x - \frac{\pi}{4} = \frac{\pi}{2} + k\pi$$

Dividing by π and solving for x :

$$x - \frac{1}{4} = \frac{1}{2} + k \implies x = k + \frac{3}{4}$$

For x to be strictly positive, we need $k \geq 0$. The first ten positive x -intercepts correspond to $k = 0, 1, 2, \dots, 9$. The sum of these arithmetic intercepts is:

$$\sum_{k=0}^9 \left(k + \frac{3}{4} \right) = \frac{9 \times 10}{2} + 10 \left(\frac{3}{4} \right) = 45 + \frac{15}{2} = \frac{105}{2}$$

5. Find the smallest positive real number x that satisfies the equation
 $\sec \left(3x - \frac{\pi}{6} \right) = \csc \left(3x - \frac{\pi}{6} \right).$

Rewrite the equation in terms of sine and cosine:

$$\frac{1}{\cos \left(3x - \frac{\pi}{6} \right)} = \frac{1}{\sin \left(3x - \frac{\pi}{6} \right)}$$

This equates to $\sin \left(3x - \frac{\pi}{6} \right) = \cos \left(3x - \frac{\pi}{6} \right)$, which simplifies to:

$$\tan \left(3x - \frac{\pi}{6} \right) = 1$$

The general solutions are given by:

$$3x - \frac{\pi}{6} = \frac{\pi}{4} + k\pi$$

Solving for x :

$$3x = \frac{5\pi}{12} + k\pi \implies x = \frac{5\pi}{36} + \frac{k\pi}{3}$$

To find the smallest positive x , we check integers k . For $k = 0$, $x = \frac{5\pi}{36} > 0$. For $k = -1$, $x = \frac{5\pi}{36} - \frac{12\pi}{36} = -\frac{7\pi}{36} < 0$. Thus, the smallest positive real number is:

$$x = \frac{5\pi}{36}$$

6. Determine the fundamental period of the function

$$f(x) = |\tan x| + |\cot x|.$$

We can algebraically manipulate $f(x)$ by expressing it entirely in terms of sine and cosine:

$$f(x) = \frac{|\sin x|}{|\cos x|} + \frac{|\cos x|}{|\sin x|} = \frac{\sin^2 x + \cos^2 x}{|\sin x \cos x|} = \frac{1}{|\sin x \cos x|}$$

Applying the double-angle identity $\sin(2x) = 2 \sin x \cos x$, we get:

$$f(x) = \frac{2}{|\sin(2x)|}$$

The fundamental period of the standard sine wave $\sin(x)$ is 2π . Compressing the graph by a factor of 2 to $\sin(2x)$ makes the period π . Finally, taking the absolute value of the sine function neatly folds the negative lobes upward, which halves the period again. Thus, the fundamental period of $|\sin(2x)|$ —and consequently of $f(x)$ —is:

$$\frac{\pi}{2}$$